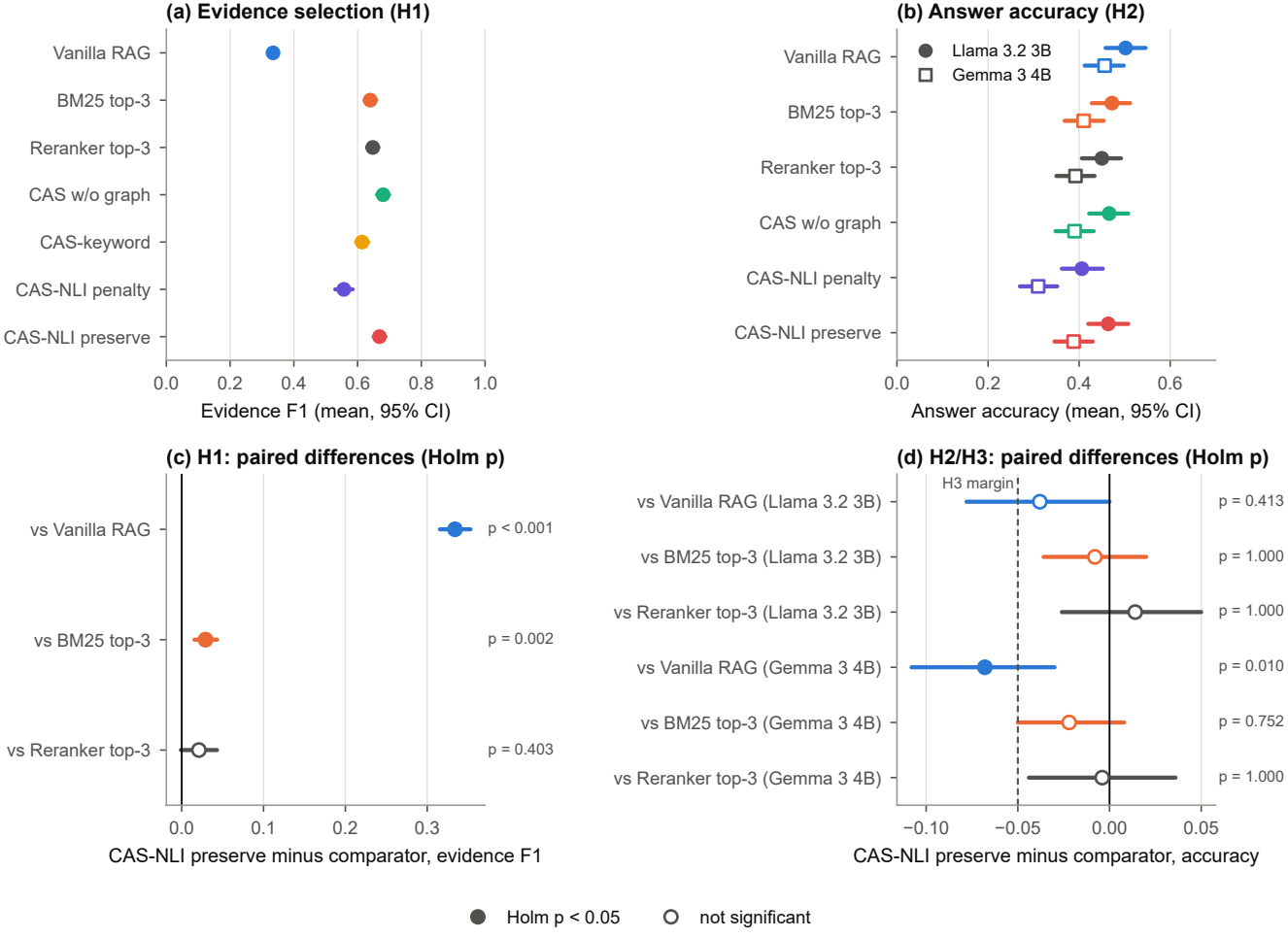

(a) Evidence selection (H1)
Vanilla RAG
BM25 top-3
Reranker top-3
CAS w/o graph
CAS-keyword
CAS-NLI penalty
CAS-NLI preserve
0.0
0.2
0.4
0.6
0.8
1.0
Evidence F1 (mean, 95% CI)
(b) Answer accuracy (H2)
Llama 3.2 3B
Gemma 3 4B
Vanilla RAG
BM25 top-3
Reranker top-3
CAS w/o graph
CAS-NLI penalty
CAS-NLI preserve
0.0
0.2
0.4
0.6
Answer accuracy (mean, 95% CI)
(c) H1: paired differences (Holm p)
vs Vanilla RAG
vs BM25 top-3
vs Reranker top-3
p < 0.001
p = 0.002
p = 0.403
0.0
0.1
0.2
0.3
CAS-NLI preserve minus comparator, evidence F1
(d) H2/H3: paired differences (Holm p)
H3 margin
vs Vanilla RAG (Llama 3.2 3B)
vs BM25 top-3 (Llama 3.2 3B)
vs Reranker top-3 (Llama 3.2 3B)
vs Vanilla RAG (Gemma 3 4B)
vs BM25 top-3 (Gemma 3 4B)
vs Reranker top-3 (Gemma 3 4B)
p = 0.413
p = 1.000
p = 1.000
p = 0.010
p = 0.752
p = 1.000
−0.10
−0.05
0.00
0.05
CAS-NLI preserve minus comparator, accuracy
Holm p < 0.05
not significant